

Ajeet Krishnasamy

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EXPERIENCE

Systems Engineer, Medical Equipment (Intern)

Oct 2020 – Sept 2021

CenTrak

Newtown, PA, United States

- Improved device reliability assessments by developing custom Python scripts and Excel models to analyze real-time medical device performance data, ensuring higher precision in monitoring systems.
- Increased maintenance planning efficiency by building automated technical performance reports for Connect Pulse™, providing engineering and operations teams with actionable insights to reduce system downtime.
- Optimized hardware lifecycle management by interpreting battery discharge and device datasets with Python (Pandas, NumPy) to identify failure trends, enabling data-driven decisions that prevented unexpected equipment outages.
- Streamlined troubleshooting workflows by structuring large-scale RTLS datasets and applying Python-based data-cleaning techniques to generate actionable findings that accelerated technical support resolutions.

Systems Engineer, Medical Equipment (Intern)

July 2019 – Sept 2019

CenTrak

Newtown, PA, United States

- Enhanced Real-Time Location Services (RTLS) functionality by applying Python-based regression and classification techniques to performance data, identifying system bottlenecks and refining location accuracy.
- Increased system validation accuracy by executing rigorous on-site testing protocols and coordinating cross-functional feedback loops with engineering teams to bridge the gap between lab results and field performance.
- Streamlined stakeholder decision-making processes by synthesizing complex technical datasets into actionable findings and formal recommendations using Python and Excel, ensuring data-driven alignment on system upgrades and feature rollouts.

PROJECTS

Wearable Hip Protector (Capstone Project) | *SolidWorks, MATLAB*

- Designed a wearable hip-protection system to mitigate impact forces from lateral and posterolateral falls in older adults.
- Conducted force analysis and ANSYS finite element stress simulations on padding layers to optimize impact absorption under dynamic loading.
- Developed a MATLAB GUI for product parameterization that allows users to input body measurements to auto-calculate pad sizing and strap geometry.
- Prototyped multilayer padding architecture using 3D printing, iterating on fit and flexibility through hands-on testing cycles.
- Designed an adjustable harness-style attachment concept with specialized waist and thigh straps for secure anatomical alignment.

3D Orbital Trajectory Animator | *Python*

- Solved Kepler's equation for two-body orbital mechanics using PyAstronomy's KeplerEllipse and animated satellite trajectories with comet trails and velocity vectors via Matplotlib's FuncAnimation and PillowWriter.
- Built an interactive orbital parameter explorer with ipywidgets and visualized four orbit regimes (LEO, GEO, Polar, Molniya) with a wireframe Earth, demonstrating critical inclination and apogee dwell.

Airfoil Generator Tool | *Python, FreeCAD, OpenFOAM*

- Engineered a reactive Python GUI that utilizes bi-directional constraint solving to mathematically synchronize NACA 4-digit geometry, chord, and aspect ratio in real-time.
- Developed a FreeCAD Python macro that automates the generation of 3D lofted wings and exports standardized STL libraries, ready for CFD analysis.
- Reduced setup time for iterative aerodynamic studies by 80% through automated mesh generation and parameter sweeping.

TECHNICAL SKILLS

Data Analysis: Python (NumPy, Pandas, SciPy, Matplotlib), MATLAB, Excel

CAD & Simulation: SolidWorks, ANSYS, FreeCAD, Fusion 360

Manufacturing: CNC Machining, 3D Printing, Lathe & Mill Operations, MIG Welding

EDUCATION

University of Ottawa

Bachelor of Applied Science (BASc), Biomedical Mechanical Engineering

Ottawa, ON, Canada

May 2026